Data Sheet

EXPERIMENT NO. 09

Title: - Simulation of Logarithmic and Anti-Logarithmic Amplifier using Op-Amp (using Tinkercad / Proteus software circuit).

Instrument & Component Required: -

Sl. No.	Apparatus / Component	Specification	Quantity
1	Op-Amp	IC 741	1
2	Diode	1N4007	1
3	Resistor	10 KΩ	1
4	DC voltage Source	15V	1
5	DC Power Supply	Variable	1
6	DC voltmeter	Digital	2

Important Formulae: -

Logarithmic Amplifier (diode/transistor in feedback path): -

The diode current, $I_D = I_S\,(e^{(V_i/V_T)} - 1) \approx I_S \cdot e^{(V_i/V_T)}$ for $V_D \gg V_T$

Since the input current $I = V_i / R$ flows through the diode (virtual ground at op-amp input):

$V_O = -V_D = -V_T \cdot \ln(V_i / (R \cdot I_S))$

- V_T = thermal voltage (≈ 26 mV at room temperature)
- I_S = diode reverse saturation current
- R = input resistor, V_i = input voltage

Anti-Logarithmic (Exponential) Amplifier (diode/transistor at input): -

$I_D = I_S \cdot e^{(V_i/V_T)}$

$Vo = -R_f \cdot I_D = -R_f\, I_S \cdot e^{(V_i/V_T)}$

- R_f = feedback resistor, V_i = input voltage applied across the diode

Observation Table – Logarithmic Amplifier

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Title:- Simulation of Logarithmic and Anti-Logarithmic Amplifier Using OP Amp (Using Tinkercad / Proteus).

Theory:-

Logarithmic Amplifier circuit utilizes a diode or Transistor in the feedback path. The output voltage is proportional to the natural logarithm of the input voltage. The diode current is I_D.

$I_D = I_S\,(e^{V_D/V_T} - 1) \approx I_S\, e^{V_D/V_T}$ $(V_D >> V_T)$.

$V_O = -V_D = -V_T \cdot \ln(V_i/(R \cdot I_S))$. $(I = V_i/R)$.

Anti logarithmic Amplifier circuit uses a diode or transistor at the input. It performs the inverse operation where the output is proportional to the antilog of the input.

The diode current $I_D = I_S \cdot e^{V_i/V_T}$.

The output voltage is $V_O = -R_f \cdot I_D = -R_f \cdot I_S \cdot e^{V_i/V_T}$

Component Required:-

		Quantity
1.	Op-Amp (IC 741)	1
2.	Diode (IN4148)	1
3.	Resistor (10KΩ)	1
4.	DC voltage (15V)	1
5.	DC power Supply (variable)	1
6.	DC voltmeter (Digital Display)	2

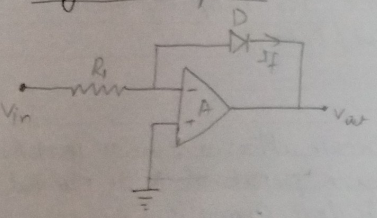
Observation Table:- (Logarithmic Amplifier)

S.N	Input Voltage (V_i) (V)	Output Voltage (V_O) (Measured)	V_O (Calculated) (V)
1.	0.5	−0.40	−0.48
2.	1	−0.51	−0.58
3.	2	−0.53	−0.52
4.	3	−0.55	−0.54

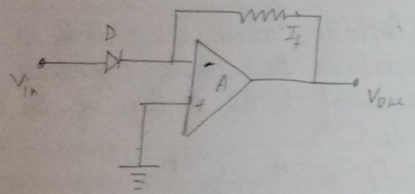
Teacher's Signature

Circuit Diagram:-

⊙ Logarithmic Amplifier:-



⊙ Anti logarithmic Amplifier:-



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■ Observation Table:- (Anti logarithmic Amplifier)

S.N	Input voltage V_i (V)	Output voltage (V) (measured)	V_o (calculated) (V)
1.	0.5	- 0.15	- 0.14
2.	0.55	- 1.13	- 1.10
3.	0.60	- 8.32	- 8.20
4.	0.62	- 18.45	- 12.10

■ Calculation:-

⊙ Log Amp: $V_o = -V_T \cdot \ln(V_i/(R_1 I_S))$

$= -0.026 \cdot \ln(0.5/(R_1 I_S))$

$= -0.48$ V

⊙ Anti log Amp: $V_o = -R_f \cdot I_S \cdot e^{V_i/V_T}$

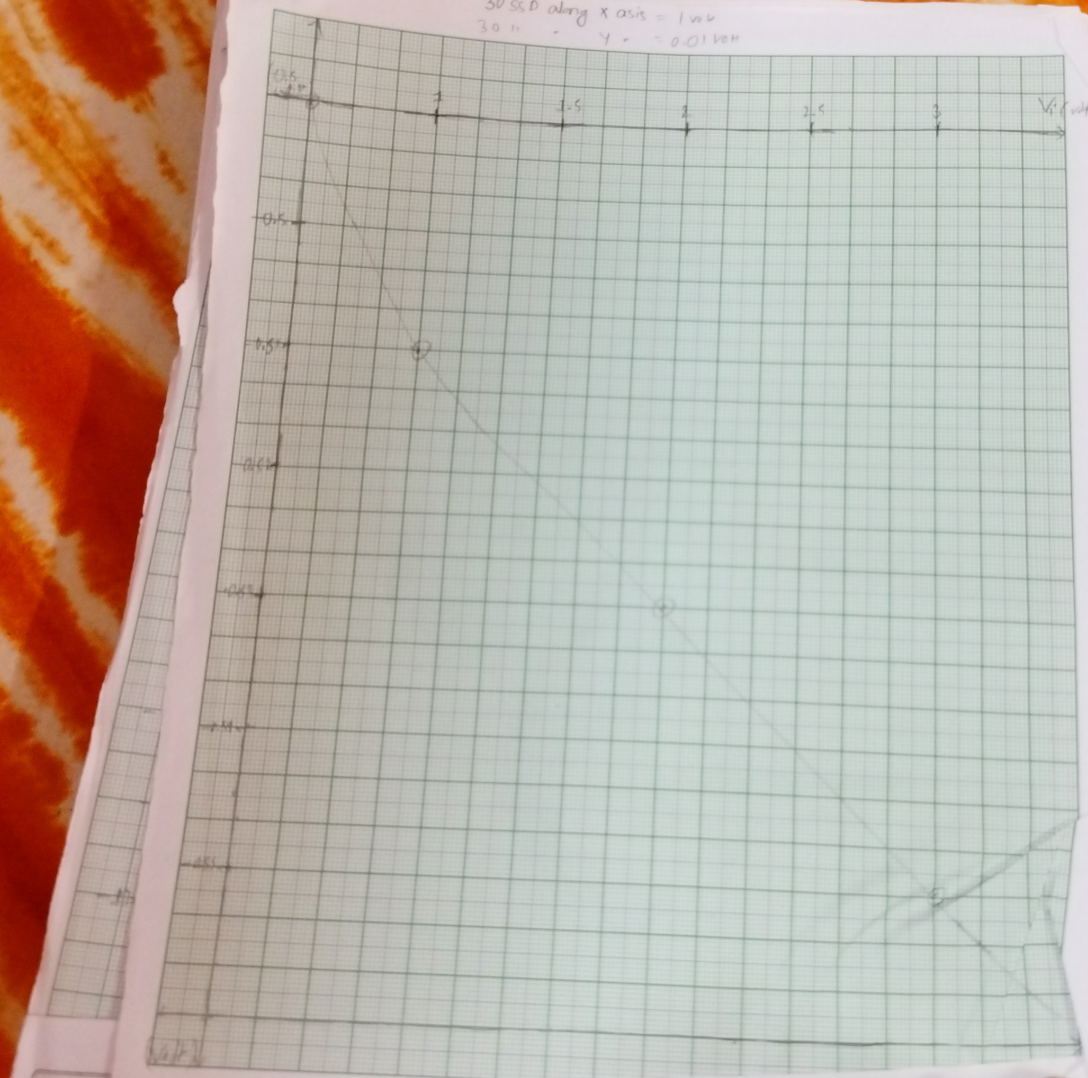
$= -R_f \cdot I_S \cdot e^{0.5/0.026} = -0.14$ V.

■ Conclusion:-

The Simulation Confirms the theoretical behavior of logarithmic and anti-logarithmic amplifier Successfully Compressed the input voltage range into a much Smaller, logarithmically Scaled negative output voltage range. Conversely, the anti-logarithmic amplifier expanded a Small input range into a rapidly increasing exponentially Scaled output (0.15V to -18.45V). The measured values closely track the theoretically Calculated values verifying the diode current equations and the virtual ground principle.

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30 SSD along X axis = 1 volt
30 " " Y " = 0.01 volt



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■ Observation Table :- (Anti logarithmic Amplifier)

S.N	Input voltage V_i (V)	Output voltage (V) (measured)	V_o (c
1.	0.5	- 0.15	
2.	0.55	- 1.13	
3.	0.60	- 8.32	
4.	0.62	- 18.45	

■ Calculation :-

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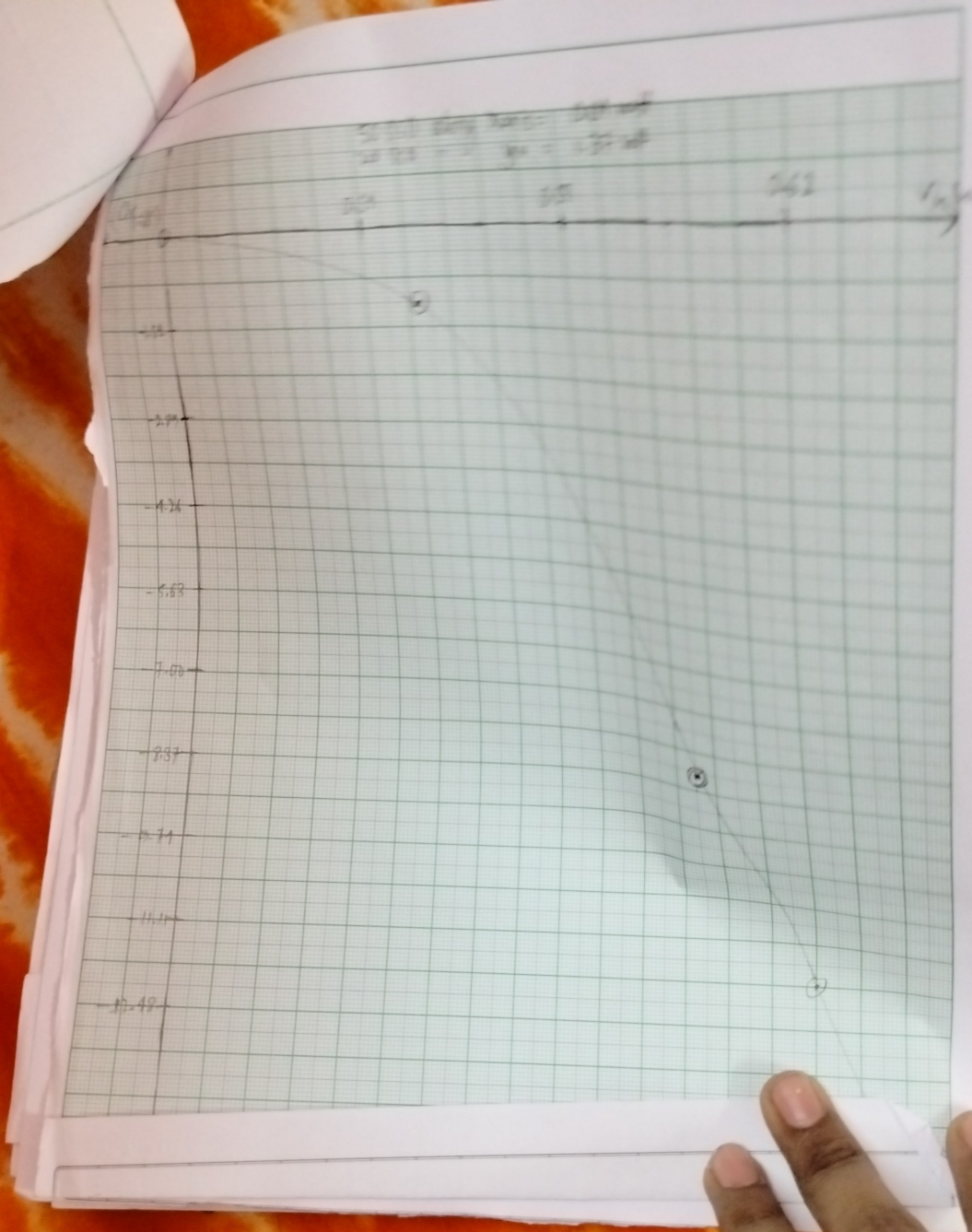
$= -0.48$ V

⊙ Anti log Amp: $V_o = -R_f \cdot I_s \cdot e^{V_i/V_T}$

$= -R_f \cdot I_s \cdot e^{0.5/0.0}$

■ Conclusion :-

The Simulation Confirms
logarithmic and anti-logarithmic amp
the input voltage range into a muc
Scaled negative output voltag range
logarithmic amplifier expanded a
rapidly increasing exponentially
The measured values closely
values verifying the diode cu
virtual ground principle



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■ Observation Table :- (Anti logarithmic Amplifier)

S.N	Input voltage V_i (V)	Output voltage (V) (measured)
1.	0.5	- 0.15
2.	0.55	- 1.13
3.	0.60	- 8.32
4.	0.62	- 18.45

■ Calculation :-

① Log Amp: $V_0 = -V_T \cdot \ln(V_i/(R_1 I_S))$

$= -0.026 \cdot \ln(0.5/(R_1$

$= -0.48$ V

② Anti log Amp: $V_0 = -R_f \cdot I_S \cdot e^{V_i/V_T}$

$= -R_f \cdot I_S \cdot e^{-0.5/0}$

■ Conclusion :-

The Simulation Confirms
logarithmic and anti-logarithmic amp
-the input voltage range into a mu
Scaled negative output voltage range
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